

JCARS SALES PERFORMANCE REPORT

1. Executive Summary

This project involved analysing a JCARS Logistics vehicle sales dataset using Microsoft Power BI. The main aim was to take the raw dataset, identify the data-quality problems in it, clean the data, and then use Power BI to create a dashboard that could help understand the company's sales and profitability performance.

The original dataset had 276 transaction records and 32 columns. During the data-cleaning process, I found different issues such as duplicate orders, inconsistent category names, missing values, invalid numbers, incorrect dates, different ways of writing payment and delivery statuses, and different currency formats. I used Power Query to review and clean these issues while trying to avoid changing values when there was no reliable reason to do so.

After the cleaning process, the final dataset contained 255 records and 34 columns. The additional columns were mainly used during the cleaning and validation process. I also created a separate date table in Power BI to support time-based analysis.

One of the main challenges was dealing with monetary values because the dataset contained KES/KSh as well as foreign currencies such as USD, EUR and ZAR. Following the project requirements, values without an explicit currency were treated as KES, while identified foreign-currency values were converted to KES using one consistent set of exchange rates. This allowed the monetary values to be analysed on the same basis.

After cleaning and validating the data, I created simple DAX measures for total orders, units sold, revenue, cost, gross profit, gross profit margin and average selling price among others. These measures were then used to build a one-page management dashboard.

The final dashboard shows that JCARS recorded approximately KSh1.24 billion in revenue from 415 units across 255 orders. However, the calculated gross profit was approximately -KSh103.27 million, giving a gross profit margin of approximately -8.34% based on the cost definition used in this project.

One of the main findings was that vehicle costs were already higher than recorded revenue before delivery and logistics costs were added. Delivery and logistics costs increased the overall loss further. The dashboard also showed differences in revenue and profitability between branches and vehicle types.

Overall, this project helped me understand how data cleaning affects the results of analysis. It also gave me practical experience in using Power Query, data modelling, DAX and Power BI visualizations to move from a messy dataset to a dashboard that can be used to investigate business performance.

2. Introduction and Business Context

JCars Logistics is a vehicle business that sells and delivers vehicles to customers across Kenya. The company records information about sales, customers, vehicles, branches, payments, deliveries, costs, returns, and customer experience.

The purpose of this project was to use Power BI to turn the raw JCars dataset into useful business information. I focused on cleaning the data, standardizing the information, analysing sales and profitability, and creating a dashboard that can help management understand the company's performance and areas that may need further investigation.

3. Dataset and Data Grain

The original dataset contained 276 transaction records and 32 columns. Each row represents one order, with information about the customer, vehicle, location, sales representative, payment, delivery, costs, and revenue.

I used Order ID as the main identifier when checking the data for duplicate orders. After cleaning and removing duplicate records, the final dataset contained 255 records and 34 columns. The additional columns were created mainly for data cleaning and validation purposes.

The dataset was initially provided as a flat table in the CSV format, which I prepared for analysis using Power Query before building the Power BI model.

4. Data Quality Audit

Before cleaning the dataset, I reviewed the columns and values to identify problems that could affect the analysis. I found several issues involving duplicates, inconsistent categories, missing values, invalid numerical values, dates, currencies, and unusual business values.

The main issues identified were:

- i. **Duplicate Order IDs:** Some orders appeared more than once, which could overstate sales and costs.
- ii. **Inconsistent Customer Types:** Similar customer categories were written in different ways, such as Company and Corporate, Individual and Person.
- iii. **Invalid Customer Ages:** Values such as 0 and -5 were not reasonable customer ages.
- iv. **Inconsistent Location Names:** Some regions and cities had alternative names, such as Mombasa and Nyali, Nyanza and Kisumu.
- v. **Inconsistent Vehicle Information:** Car makes, models, fuel types, and transmission values contained spelling and formatting variations.

- vi. **Invalid Vehicle Year:** The value 202A was not a valid year.
- vii. **Inconsistent Units Sold:** Some quantities were written as text, such as two, one, and 3 cars.
- viii. **Invalid Numerical Values:** Some fields contained negative values where they were not reasonable, such as negative costs and ratings.
- ix. **Inconsistent Payment and Delivery Statuses:** Similar statuses were written using different formats and spellings.
- x. **Inconsistent Customer Ratings:** Ratings appeared as numbers, text, and formats such as 4/5 and 4 out of 5. Some even higher than the maximum rating score, 5. Like 6.
- xi. **Mixed Currencies:** Monetary values were recorded using KES/KSh, USD, EUR, and ZAR, while some values had no currency indicated.
- xii. **Inconsistent Date Formats:** Order and delivery dates appeared in different formats, including Excel serial numbers, text dates, and invalid dates, like April 31.
- xiii. **Unusual Revenue Valuea:** Some revenue values were negative, zero, or did not match the expected revenue calculation.

I did not automatically change every unusual value. Where there was not enough evidence to determine the correct value, I retained it or treated it as missing and documented the decision.

5. Data Cleaning and Transformation

I used Power Query in Power BI to clean and prepare the dataset for analysis. I worked through the columns and standardized inconsistent values, corrected formatting issues, handled missing and invalid values, removed duplicate orders, and converted columns to appropriate data types.

Some of the main cleaning activities included standardizing customer types, locations, vehicle information, payment and delivery statuses, units sold, customer ratings, monetary values, and dates.

Where a value could not be corrected confidently, I retained it as an anomaly or treated it as missing instead of making an unsupported correction. After cleaning, the dataset was reduced from 276 to 255 transaction records and prepared for validation and analysis.

6. Currency Standardization

The dataset contained monetary values in KES/KSh, USD, EUR and ZAR, as well as values without a currency indicator and some with a question mark “?”. I treated values without an explicit currency as KES, as required by the assessment.

I converted the identified foreign-currency values to KES using one consistent set of exchange rates:

- i. 1 USD = 129.64 KES
- ii. 1 EUR = 147.71 KES
- iii. 1 ZAR = 7.98 KES

The rates were obtained from Google Finance and referenced on 27 September 2026, 02:58 EAT. The converted values were then used for the financial analysis.

7. Data Validation

After cleaning, I validated the dataset before using it in Power BI. I checked the number of records, duplicate Order IDs, data types, dates, monetary fields, and important calculated values.

The final dataset contained 255 records and 34 columns, with no duplicate Order IDs. Order Date and Delivery Date were converted to Date type, while the main monetary fields were converted to numeric KES currency values.

I also created an expected revenue calculation to compare against the recorded revenue. Where differences occurred, I investigated them rather than automatically changing the recorded revenue.

8. Data Modelling

The cleaned JCars dataset remained as the main transaction table in Power BI. I also created a separate DimDate table to support time-based analysis.

The DimDate table was connected to the transaction table using the Order Date field with a one-to-many relationship. This allowed me to analyse revenue and other measures over time while keeping the model simple.

9. Business Logic and DAX

After cleaning and modelling the data, I created DAX measures to calculate the main business performance indicators used in the report. I used measures instead of relying only on automatic Power BI aggregations so that the calculations could be reused across different visuals and filters.

a. Total Orders

Total Orders = `COUNTROWS(Jcars_data_working)`

This counts the number of transaction records in the cleaned dataset.

b. Total Units Sold

Total Units Sold = `SUM(Jcars_data_working[Units Sold])`

This calculates the total number of vehicles sold.

c. Total Revenue

Total Revenue = `SUM(Jcars_data_working[Revenue Recorded (KES)])`

This calculates the total recorded revenue using the standardized KES revenue field.

d. Total Cost

Total Cost = `SUMX(Jcars_data_working, (Jcars_data_working[Unit Cost (KES)] * Jcars_data_working[Units Sold]) + Jcars_data_working[Delivery Fee (KES)] + Jcars_data_working[Logistics Cost (KES)])`

I defined total cost as the vehicle cost based on units sold, plus delivery and logistics costs.

e. Gross Profit

Gross Profit = `[Total Revenue] - [Total Cost]`

This calculates the difference between recorded revenue and the total cost defined for this project.

f. Gross Profit Margin

Gross Profit Margin = `DIVIDE([Gross Profit], [Total Revenue])`

This shows gross profit as a percentage of revenue.

g. Average Selling Price

Average Selling Price = `DIVIDE(SUMX(Jcars_data_working, Jcars_data_working[Unit Selling Price (KES)] * Jcars_data_working[Units Sold]), [Total Units Sold])`

This calculates the average selling price per vehicle based on the selling price and units sold.

h. Return Rate

Return Rate = `DIVIDE([Returned Orders], [Total Orders])`

These measures were used to identify the number and percentage of orders marked as returned.

i. Average Delivery Days

Average Delivery Days = `AVERAGEX(Jcars_data_working, DATEDIFF(Jcars_data_working[Order Date], Jcars_data_working[Delivery Date], DAY))`

This calculates the average number of days between the order date and delivery date.

These measures provided the calculations used in the dashboard and helped me analyse revenue, costs, profitability, sales volume, returns, and delivery performance.

10.Business Questions

The analysis was guided by the business questions in the assessment. I grouped related questions together and used the dashboard visuals and DAX measures to answer the main areas that could be analysed from the available data.

The main questions addressed were:

- What are the total revenue, orders, and units sold?
- What is the overall gross profit and gross profit margin?
- How does revenue change over time?
- How does revenue differ between branches?
- How does revenue differ between vehicle types?
- Which vehicle types generate positive or negative gross profit?
- Which branches generate positive or negative gross profit?
- How do average vehicle selling prices compare with unit costs across vehicle types?
- Which areas of the business may require further investigation based on their revenue and profitability?

The dashboard was designed to provide a high-level view of these questions using KPI cards, trend analysis, branch and vehicle comparisons, profitability visuals, and slicers for Vehicle Type and Branch.

This is much better for your report because it connects Section 10 directly to the dashboard you actually created rather than claiming that you answered every question in the assessment brief.

11.Analyst-Defined Business Questions

During the analysis, I came up with a few additional questions:

1. Are vehicle costs higher than the revenue from vehicle sales?
2. Which vehicle types make a profit or a loss?
3. Which branches make a profit or a loss?
4. How do delivery and logistics costs contribute to the overall cost?
5. How do vehicle selling prices compare with their costs?

These questions helped me look beyond total sales and understand the factors affecting the company's profitability.

12. Investigation and Exceptions

I investigated unusual values and patterns rather than automatically treating them as errors. Some examples included negative revenue, unusual vehicle years, zero values, and differences between recorded revenue and expected revenue.

The profitability analysis also showed that some vehicle types and branches had negative gross profit while others were positive. These differences were treated as areas for further investigation rather than assuming that the data was incorrect.

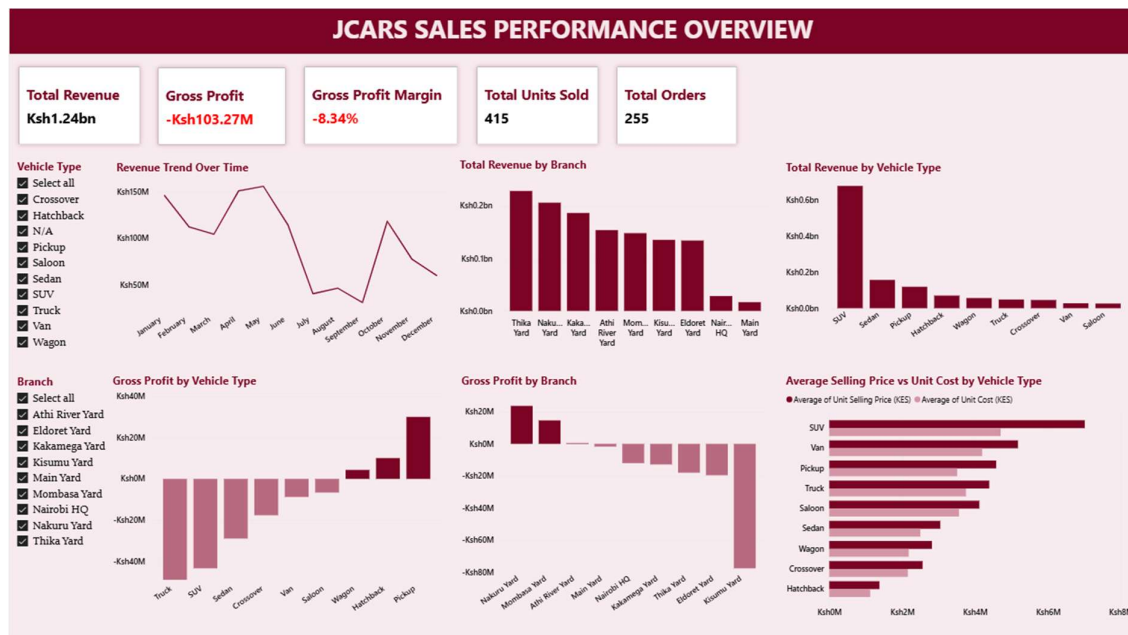
One important finding was that total vehicle cost was higher than recorded revenue before delivery and logistics costs were added. This suggested that the overall loss was not caused only by delivery and logistics expenses.

13. Executive Dashboard

I created a one-page Executive Dashboard titled JCARS SALES PERFORMANCE OVERVIEW. The dashboard was designed to give management a quick view of sales, revenue, costs, and profitability.

The dashboard includes five KPI cards showing total revenue, gross profit, gross profit margin, units sold, and total orders. It also includes revenue trends, revenue by branch, revenue by vehicle type, gross profit by vehicle type, gross profit by branch, and a comparison of average selling price and unit cost.

I also added Vehicle Type and Branch slicers to allow users to explore the results interactively.



14.Detailed Analytical Report

The detailed analysis was designed to investigate the patterns shown in the Dashboard in more detail. I focused mainly on revenue, vehicle performance, branch performance, profitability, and cost differences. The visuals were selected based on the questions they helped answer rather than creating a separate visual for every business question. This allowed the rep

15.Dashboard and Report Analytical Flow

The Dashboard provides the main overview of the business. I used it to identify differences in revenue and profitability across branches and vehicle types, as well as changes in revenue over time. The dashboard can then be used to identify areas that need further investigation, especially vehicle types and branches with negative gross profit.

16.Interactivity

I added Vehicle Type and Branch slicers to the dashboard. These allow the user to filter the visuals and compare the performance of different vehicle types and branches. The slicers make it easier to explore the results without creating separate dashboards for each category.

17.Management Insights

From the analysis, I identified the following main insights:

- i. JCARS recorded about KSh1.24 billion in revenue from 415 units across 255 orders.
- ii. The overall gross profit was approximately -KSh103.27 million, giving a gross profit margin of -8.34%.
- iii. Vehicle costs were already higher than revenue before delivery and logistics costs were included.
- iv. Profitability varied across vehicle types, with some types generating positive gross profit while others generated losses.
- v. Branch performance also differed, with some branches showing positive gross profit and others showing negative gross profit.
- vi. The revenue trend changed throughout the year, with noticeable increases and decreases across different months.
- vii. The selling price compared with unit cost showed that some vehicle types had a relatively small difference between selling price and cost, which may contribute to their lower profitability.

18.Management Recommendations

Based on the findings, I recommend that management:

- i. Review vehicle pricing and costs, especially for vehicle types showing negative gross profit.
- ii. Investigate branches with negative profitability to understand the differences in costs, sales and revenue.
- iii. Review delivery and logistics costs and identify areas where these costs can be controlled without affecting service quality.
- iv. Monitor vehicle profitability regularly instead of focusing only on revenue and units sold.

These recommendations are based on the patterns identified in the analysis and should be investigated further before major business decisions are made.

19.Assumptions and Business Rules

I made several assumptions during the cleaning and analysis process:

- i. Monetary values without a currency indicator were treated as KES.
- ii. Foreign currencies were converted to KES using the selected exchange rates.
- iii. Missing or invalid values were treated as missing where they could not be corrected confidently.
- iv. Duplicate Order IDs were removed because each row was treated as one order.
- v. Negative values were not automatically changed unless there was a clear reason to do so.
- vi. The Revenue Recorded field was used as the recorded revenue for the main analysis.
- vii. Total cost was calculated using vehicle cost + delivery fee + logistics cost.
- viii. Gross profit was calculated as revenue – total cost.
- ix. 202A was treated as an invalid vehicle year, while unusual years such as 1899 and 2032 were retained because there was no reliable information to correct them.
- x. Ambiguous dates were interpreted using the available date information and converted to a consistent Date format.

20.Challenges and What I Learned

One of the main challenges was cleaning the dataset because many columns contained different formats and inconsistent values. I also found the currency conversion and mixed date formats challenging.

Working through these problems helped me understand the importance of checking data before analyzing it. I also learned how Power Query can be used to clean real-world data and how DAX measures can turn the cleaned data into useful business information.

Building the dashboard also helped me understand that good analysis is not only about creating charts. The visuals need to answer business questions and make unusual patterns easier to identify.

21.Conclusion

This project took the JCars dataset from a raw and inconsistent state to a cleaned dataset that could be analysed in Power BI. I used Power Query for cleaning, created a simple data model and DAX measures, and built an interactive Executive Dashboard.

The analysis showed that although JCARS generated significant revenue, the business had an overall negative gross profit based on the cost definition used in this project. The differences between vehicle types and branches also showed areas that could be investigated further.

Overall, the project gave me practical experience in data cleaning, Power Query, data modelling, DAX, Power BI visualisation, and business analysis.